PART I: FOUNDATIONS

Topic 6: Missing Time Points and Reporting Gaps

The Core Concept

Time series analysis relies on a **continuous chain** of observations. A missing link does not just create a visual gap — it creates an unanswered question about what actually happened during that period.

Why It Matters

In public safety data, gaps commonly occur due to system upgrades, paper backlogs, or shifts in reporting standards. If we simply connect the dots across a gap, we are guessing — and our guess could be dangerously wrong.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
Jan	12	25	400
Feb	10	22	380
Mar	14	28	450
Apr	15	24	460
May	18	26	550
Jun	22	24	600
Jul	25	27	700
Aug	**(Missing)**	23	680
Sep	19	22	500
Oct	16	**85 ★ Outlier**	420
Nov	14	24	390
Dec	11	26	350

Precinct North is missing data for August. If we connect July (25) directly to September (19), we implicitly assume August was around 22. But consider the alternatives:

- **If a major protest occurred in August:** the real number could have been 100.
- **If the precinct was closed for renovation:** the real number could have been 0.

Before treating a gap as a smooth transition, always ask:

1. Why is this data missing? (System failure? Policy change? Staffing gap?)
2. Was the gap random, or does it coincide with a known event?
3. Can we recover the data from another source?

💡 **The Insight:** A "0" is a data point. A "Missing" is a question mark. Treat them very differently.

Key Takeaway: Never assume a gap is smooth. Investigate why the data is missing before trusting any trend that spans it.